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ASSIGNMENT 1: KNOWLEDGE REPRESENTATION

THEME: SMART EDUCATION

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Knowledge Representations

1. Student Knowledge Profile

This knowledge representation stores every student's proficiency in every topic. It helps the system to recognise weak and strong areas, and detect knowledge gaps. This KR is important for personalized learning, as it produces the core information for the AI to decide what kind of learning resources needed for the students.

i) *IF x is in the student database, THEN x is a student*

ii) *IF y is listed in the course structure, THEN y is a topic.*

iii) *IF $\text{Proficiency}(x,y,l)$ is true, THEN student x has proficiency score of l in topic y .*

iv) *IF there exist a proficiency level l such that $\text{Proficiency}(x,y,l)$ is true AND l is less than 40, THEN x is weak in topic y*

v) *IF there exist a proficiency level l such that $\text{proficiency}(x,y,l)$ is true AND is more than 80, THEN x is strong in topic y .*

vi) *IF a student is not weak AND not strong in a topic, AND their proficiency falls between 40 and 80, THEN x is moderate in topic y .*

Explanation:

The student knowledge profile evaluates each student's understanding of every topic by assigning a proficiency score. If the score is below than 40, the student is marked as weak in that topic, showing that this student needs additional guidance. When the score meets or exceeds 80, the student is categorized as strong, indicating that the student is ready for advanced materials. And the student is marked as moderate in that topic if the score falls between 40 and 80. By identifying the proficiency levels for every student, the system can accurately analyze learning gaps and prepare a suitable recommendation such as practice, modules, or learning videos.

Predicate Logic/First Order Logic:

i) $\forall x (\text{InStudentDatabase}(x) \rightarrow \text{Student}(x))$

ii) $\forall y (\text{InCourseStructure}(y) \rightarrow \text{Topic}(y))$

iii) $\forall x \forall y \forall l (\text{Proficiency}(x,y,l) \rightarrow \text{Student}(x) \wedge \text{Topic}(y))$

iv) $\forall x \forall y (\text{Weak}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge l < 40))$

v) $\forall x \forall y (\text{Strong}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge l \geq 80))$

vi) $\forall x \forall y (\text{Moderate}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge 50 \leq l < 80))$

For every x that exists in the student database, then x is a student.

For every y that exists in course structure, then y is a topic.

For every student x , topic y , and every level l , if x has proficiency level l for topic y , then x must be a student and y must be a topic.

For every student x and every topic y , x is weak in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is less than 50.

For every student x and every topic y , x is strong in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is more than 80.

For every student x and every topic y , x is moderate in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is greater or equal to 50, but less than 80.

2. Personalize Learning Path

It makes predictions about the learning resources based on the student knowledge profile that considers their skills, the areas where they are lacking, and their grasp of the basic topics so that their development is at the right speed and the right learning techniques are used.

i) *IF a student is weak in a topic THEN recommend a remedial module.*

ii) *IF a student is moderate in a topic THEN recommend a practice module.*

iii) *IF a student is strong in a topic AND the student is weak in the next topic THEN recommend the next topic module.*

Explanation:

If a student has trouble with a subject, the system will first evaluate the student's knowledge and skills in related subjects before recommending the most appropriate course material. In the situation where the students have a weak understanding on the topic, this approach will offer a remedial module designed to reinforce their basic concepts. If the students have moderate proficiency, the AI system will offer them a practice module that will aid them to comprehend better and gain skills. If the student shows a strong understanding on the topic, the AI system will offer the next topic module. This approach helps each student get a personalized learning way that leads them through content as their needs and speed by smartly choosing learning materials depending on the student's ability.

Predicate Logic/First Order Logic:

i) $\forall x \forall y ((\text{weak}(x,y)) \rightarrow \text{recommend}(x, \text{remedial_module}(y)))$

ii) $\forall x \forall y ((\text{moderate}(x,y)) \rightarrow \text{recommend}(x, \text{practice_module}(y)))$

iii) $\forall x \forall y \forall z ((\text{strong}(x,y) \wedge \text{subtopic}(y,z) \wedge \text{weak}(x,z)) \rightarrow \text{recommend}(x, \text{next_topic_module}(z)))$

For every student x and every topic y , if the student x is weak in topic y then a remedial module for topic y is recommended to the student.

For every student x and every topic y , if the student x is moderate in topic y then a practice module for topic y is recommended to the student.

For every student x , every main topic y , and every related next topic z , if the student x is strong in the main topic y and weak in the next topic z , then the next topic module for y is recommended to the student.

3. Real-time Engagement Monitor

This knowledge representation tracks student engagement and attention levels while they are learning to identify the times students are losing their focus, getting confused, or not engaging well with the material so the system can intercede at the right moment to keep them on track.

i) IF a student's response time increases significantly AND answer accuracy decreases THEN alert for possible confusion or fatigue.

ii) IF a student repeatedly skips questions or leaves sections incomplete AND time spent per question is very short THEN detect disengagement and recommend motivational break.

iii) IF a student shows consistent engagement (regular progress, steady response time) AND maintains good accuracy THEN continue current learning path without interruption.

iv) IF a student's engagement suddenly drops after specific content type (video/text/exercise) THEN flag that content format as less effective for that student.

v) IF a student attempts same question multiple times AND time spent increases each attempt THEN identify struggling point and offer hint or alternative explanation.

Explanation:

The real time engagement monitor keeps track of how students are actually interacting with their given learning materials while they are working on them. This is not just about getting the answers right or wrong, the system picks up on behavioral patterns showing someone is struggling. When someone takes longer time to answer questions and at the same time the accuracy of the answers drops, that means they are either confused or losing focus. The system will catch this early and alerts changes needed before the student gets frustrated. When students start rushing through the material, skipping parts, or barely spending time before moving on to the next part, that is a clear signal that they are mentally checked out. The system will detect this disengagement trend and suggest stepping away for a bit and taking a break.

The system also notices types of content work for each student. If engagement consistently drops right after watching videos or reading text sections, the information and data will tell the system those formats are not effective for that specific student. It can then adjust recommendations to show them content in formats they respond to way better. The system

will recognize the struggles when the student keeps attempting the same question multiple times with a helpful hint or offer explanations.

Predicate Logic/First Order Logic:

- i) $\forall s \quad \forall t \quad ((\text{ResponseTimeIncrease}(s,t) \quad \wedge \quad \text{AccuracyDecrease}(s,t)) \rightarrow \text{AlertConfusionOrFatigue}(s,t))$
- ii) $\forall s \quad \forall t \quad ((\text{FrequentSkipping}(s,t) \quad \wedge \quad \text{VeryShortTimePerQuestion}(s,t)) \rightarrow (\text{DetectDisengagement}(s,t) \wedge \text{RecommendBreak}(s)))$
- iii) $\forall s \quad \forall t \quad ((\text{ConsistentEngagement}(s,t) \wedge \text{GoodAccuracy}(s,t)) \rightarrow \text{ContinueCurrentPath}(s,t))$
- iv) $\forall s \quad \forall c \quad \forall t \quad ((\text{EngagementDropAfterContent}(s,c,t)) \rightarrow \text{FlagIneffectiveFormat}(s,c))$
- v) $\forall s \quad \forall q \quad ((\text{MultipleAttempts}(s,q) \quad \wedge \quad \text{IncreasingTimePerAttempt}(s,q)) \rightarrow (\text{IdentifyStrugglingPoint}(s,q) \wedge \text{OfferHint}(s,q)))$
- vi) $\text{ResponseTimeIncrease}(s,t) \leftrightarrow \exists r1 \quad \exists r2 \quad (\text{PreviousResponseTime}(s,t,r1) \quad \wedge \quad \text{CurrentResponseTime}(s,t,r2) \wedge r2 > 1.5 \times r1)$
- vii) $\text{ConsistentEngagement}(s,t) \leftrightarrow (\text{RegularProgress}(s,t) \wedge \text{SteadyResponseTime}(s,t) \wedge \neg \text{FrequentSkipping}(s,t))$

For every student s and topic t , if the student's response time increases significantly and answer accuracy decreases simultaneously, then the system alerts for possible confusion or fatigue.

For every student s and topic t , if the student frequently skips questions and spends very little time on each question, then disengagement is detected and a motivational break is recommended.

For every student s and topic t , if the student shows consistent engagement patterns and maintains good accuracy, then the current learning path continues without interruption.

For every student s , content type c , and topic t , if engagement suddenly drops after encountering specific content type, then that content format is flagged as less effective for that student.

Response time is considered significantly increased if the current response time is more than 1.5 times the previous average response time.

Consistent engagement exists if and only if the student demonstrates regular progress, maintains steady response time, and does not frequently skip content.

4. Adaptive Difficulty Progression Tracker

This knowledge representation monitors and adjusts the difficulty level of learning materials and assessments in real-time based on each student's performance path, ensuring struggling students receive appropriately suitable content while advanced students are continuously challenged with harder materials to prevent boredom and maintain their engagement towards learning. This KR specifically focuses on adjusting the challenge level within the learning materials, working in tandem with Personalized Learning Path KR which determines what type of content the student needs.

- i) IF a student consistently scores below 40% on assessments for a topic AND shows no improvement over 3 attempts THEN reduce difficulty level and provide simplified content with guided examples.*
- ii) IF a student consistently scores above 85% on current difficulty level AND completes tasks quickly THEN increase difficulty level and introduce advanced challenge materials.*
- iii) IF a struggling student reaches 65% in previously weak topic THEN unlock intermediate-level content to bridge gap toward advanced materials.*

Explanation:

The adaptive difficulty progression tracker closes the performance quality gap between struggling and advanced students by making sure everyone works at the correct specified challenge level. Unlike the Personalized Learning Path which decides whether students need remedial, or practice, this KR determines how challenging those materials should be based on the ongoing performance trends. When students keep scoring poorly after multiple attempts, the system recognizes they need simpler content with step-by-step examples rather than repeating the same difficult material. This prevents them from falling further behind and builds their confidence with achievable goals.

For students with high scores and quick completion, the system raises difficulty and introduces complex problems to keep them engaged. This step prevents advanced students from getting bored and disengaged. The tracker also monitors students gradually improving from weak to moderate to high performance. It carefully increases difficulty in small steps aligned with their improvement scale, helping them learn without sudden risky jumps. Once struggling students reach moderate proficiency that is needed, the system unlocks intermediate content bridging them toward advanced materials. Even for advanced students, if performance drops and time increases significantly on harder content, the system backs off to find that optimal challenge

zone where work is engaging but not frustrating. This dynamic difficulty adjustment ensures that once learning path is determined, the materials presented at each stage are optimally calibrated to each student's current capability level.

Predicate Logic/First Order Logic:

- i) $\forall s \forall t ((\text{ConsistentlyLowScore}(s,t) \wedge \text{NoImprovement}(s,t,3)) \rightarrow (\text{ReduceDifficulty}(s,t) \wedge \text{ProvideSimplifiedContent}(s,t)))$
- ii) $\forall s \forall t ((\text{ConsistentlyHighScore}(s,t) \wedge \text{QuickCompletion}(s,t)) \rightarrow (\text{IncreaseDifficulty}(s,t) \wedge \text{IntroduceAdvancedMaterials}(s,t)))$
- iii) $\forall s \forall t ((\text{GradualImprovement}(s,t) \wedge \text{ConsistentEffort}(s,t)) \rightarrow \text{ProgressivelyIncreaseDifficulty}(s,t))$
- iv) $\forall s \forall t ((\text{PreviouslyWeak}(s,t) \wedge \text{CurrentlyModerate}(s,t)) \rightarrow \text{UnlockIntermediateContent}(s,t))$
- v) $\forall s \forall t ((\text{AdvancedStudent}(s) \wedge \text{DecliningPerformance}(s,t) \wedge \text{TimeIncreaseSignificant}(s,t)) \rightarrow \text{ReduceToOptimalLevel}(s,t))$
- vi) $\text{ConsistentlyLowScore}(s,t) \leftrightarrow \forall a (\text{RecentAssessment}(a,t) \wedge \text{TakenBy}(s,a) \rightarrow \text{Score}(s,a) < 50)$
- vii) $\text{ConsistentlyHighScore}(s,t) \leftrightarrow \forall a (\text{RecentAssessment}(a,t) \wedge \text{TakenBy}(s,a) \rightarrow \text{Score}(s,a) \geq 85)$
- viii) $\text{GradualImprovement}(s,t) \leftrightarrow \exists a1 \exists a2 \exists a3 (\text{Assessment}(a1,t) \wedge \text{Assessment}(a2,t) \wedge \text{Assessment}(a3,t) \wedge \text{Score}(s,a1) < \text{Score}(s,a2) < \text{Score}(s,a3))$

For every student s and topic t , if the student consistently scores below 50% and shows no improvement over 3 attempts, then difficulty is reduced and simplified content with guided examples is provided.

For every student s and topic t , if the student consistently scores above 85% and completes tasks quickly, then difficulty is increased and advanced challenge materials are introduced.

For every student s and topic t , if the student shows gradual improvement and maintains consistent effort, then difficulty is progressively increased while maintaining appropriate pace.

For every student s and topic t , if the student was previously weak but currently achieves moderate proficiency, then intermediate-level content is unlocked to bridge toward advanced materials.

A student shows consistently low scores if and only if all recent assessments for that topic result in scores below 50%.

A student shows consistently high scores if and only if all recent assessments for that topic result in scores of 85% or higher.

Gradual improvement exists if and only if there exist three sequential assessments where each subsequent score is higher than the previous one.

5. Instructor and Class Analytics

Determine Weak Subjects and Assist Lecturers in Making Decisions

i) *IF average_class_proficiency < threshold OR topic_mastery_low THEN alert_topic = on*

ii) *IF number_of_weak_students $\geq$ weak_limit THEN notify_lecturer = on*

iii) *IF topic_rank $\leq$ 3 THEN top_weak_topic = true*

iv) *IF proficiency_drop_is_large THEN decline_alert = on*

Explanation:

Instead of concentrating solely on individual students, the Instructor and Class Analytics knowledge representation models the performance of the entire class. It creates class-level indicators by combining each student's competency, assessment results and learning activity log. These indicators assist instructors in determining which subjects the majority of students find difficult and how the overall performance of the class corresponds to desired learning standards. The method can identify trends like widespread misconceptions or persistent poor scores in particular subjects by computing the average proficiency for each topic and keeping track of the number of students who exhibit weaknesses.

The system automatically sends out signals or notice to the instructor when too many students show poor performance or when the average proficiency for a topic drops below a certain threshold. These warnings indicate that the subject might need to be addressed right away such as by reteaching, adding more exercises, changing the lesson's pace or providing extra resources. Additionally, the system records performance trends over time and ranks topics from strongest to weakest, allowing instructors to discover the most problematic subjects and identify sharp drops in student comprehension which is going to guarantee proactive instructional modifications and better data driven choices to serve the entire class.

Predicate Logic / FOL

i) $\forall c \forall t (Class(c) \wedge AvgProficiency(c, t, avg) \wedge (avg < ClassThreshold \vee TopicWeak(c, t)) \rightarrow TopicAlert(c, t))$

ii) $\forall c \forall t (WeakCount(c, t, n) \wedge n \geq WeakLimit \rightarrow NotifyLecturer(c, t))$

iii) $\forall c \forall t (WeakerTopicRank(c, t, r) \wedge r \leq 3 \rightarrow TopWeakTopic(c, t))$

iv) $\forall c \forall t \forall t1 \forall t2 \quad (\text{AvgProficiencyAt}(c,t,t1,a1) \wedge \text{AvgProficiencyAt}(c,t,t2,a2) \wedge \text{DroppedSignificantly}(a1,a2) \rightarrow \text{RapidDec}$

For each class (c) and topic (t), if that topic's average proficiency falls below the class threshold ($\text{avg} < \text{ClassThreshold}$) or if ($\text{TopicWeak}(c, t)$) indicates that the topic is weak for the class, then the instructor is subsequently prompted to take action by the system's activation of $\text{TopicAlert}(c, t)$.

For every class (c), if the number of weak students in each class(c) and topic(t) exceeds or equals the weakness limit, then the instructor will be notified.

For each class(c) and the topic(t), the topic is regarded as one of the top weak topics if its weakness rating is 3 or less.

If proficiency declines considerably for each class(c), topic(t) and time periods(t1 and t2), the system initiates a rapid decline notice.

Knowledge Representation Involved to Achieve Goals

Knowledge Representation (KR) is essential to smart education because it allows AI systems to comprehend, interpret and apply data about students, learning materials and class performance. KR enables systems to offer individualised learning experiences, efficiently track progress and adjust to individual needs by explicitly expressing student's knowledge, learning paths, assessment results and scheduling limitations.

The AI system seeks to narrow the performance gap between struggling and advanced students, discover and close knowledge gaps early, personalise learning modules and let instructors track students' progress. Every KR component, from Student Knowledge Profile to Instructor and Class Analytics is made to accomplish these goals by logically and effectively organising pertinent data for decision making.

1) Student Knowledge Profile KR → Achieving Goal: Identify and Overcome Knowledge Gaps Early

Every student's competency in every subject, including their areas of strength and weakness as well as how their performance changes overtime are stored in the Student Knowledge Profile KR. The method precisely detects knowledge gaps by using structured predicates like weak, moderate and strong to characterise student abilities. This makes it possible to identify problems early on and offer prompt solutions, guaranteeing that students fill in any gaps before they fall behind and promoting long-term comprehension of the curriculum.

2) Personalized Learning Path KR → Achieving Goal: Personalize Learning Modules

By using the student knowledge profile, prerequisite relationships between topics and rules that determine whether a student needs remedial, practice or next-topic modules. The Personalized Learning Path KR finds the best learning resources for each student by automatically creating highly customised learning paths based on the demands and learning pace of each student by expressing subject dependencies, subtopic structures and logic-based conditions. This method guarantees flexible and focused learning opportunities, assisting students in strengthening their areas of weakness while successfully completing the course materials.

3) Real-time Engagement Monitor KR → Achieving Goal: Identify and Overcome Knowledge Gaps Early

This KR monitors students' attention and engagement during the learning process, spotting instances in when they get disinterested in the subject matter, lose focus, or become confused. The technology identifies problems in real time by tracking behavioural trends, such as longer reaction times with worse accuracy, skipped questions, repeated attempts, or abrupt declines in engagement following particular content categories. When such tendencies are detected, the AI can step in with suggestions, other explanations, motivational pauses, or content format modifications that better fit the student's learning preferences. This proactive strategy guarantees that difficulties are promptly resolved, keeping students from falling behind and promoting ongoing, productive learning.

4) Adaptive Difficulty Progression Tracker KR → Achieving Goal: Reduce the Gap Between Struggling and Advanced Students

In order to fit each student's current competency level, this KR dynamically modifies the difficulty of learning materials and exams based on real-time performance monitoring. While advanced students who score highly and finish tasks quickly are given more difficult materials to keep them engaged, struggling students who consistently score low or show no improvement receive simplified content with guided examples to keep them from falling further behind. The approach bridges the gap towards advanced topics by gradually increasing difficulty for pupils who consistently improve and unlocking intermediate content for those who are progressing from weak to moderate proficiency. This KR guarantees that all students advance at a suitable rate by consistently adjusting the degree of difficulty to each student's performance trajectory. This promotes fair learning outcomes and lessens gaps between struggling and advanced students.

5) Instructor & Class Analytics KR → Achieving Goal: Assist Lecturers in Tracking Each Student's Progress Effectively

Individual student performance is combined into class-level indicators via Instructor and Class Analytics KR. These indicators include average competency, weak topic rankings, broad difficulty alerts and abrupt performance reductions. Instructors or lecturers can rapidly determine which topics the class struggles with, which students require extra attention and which teaching method works best by using ideas like class averages, weak counts, topic alerts and trends over time. Consequently, the instructors are able to effectively track students' progress and make well informed decisions that direct learning, eventually promoting both student achievement and overall class performance.

Topic	AI Analysis (The "Why")	Recommended Action (The "How")
Heuristic Search	68% of students failed the "Admissibility" sub-module.	Deploy Adaptive Quiz: Re-evaluate the concept of with a visual trial exam.
Intelligent Agents	Proficiency dropped significantly this week.	Reteach Session: Release "Agent Environments" video tutorial to 12 weak students.
Search Algorithms	Topic Mastery is low despite high student effort	Resource Push: Add "Dijkstra vs. BFS" practice set to personalized learning paths.